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# SDC Hackathon

## Phase 1 : Preliminary Project Proposal

**Hosted by Xiamen University Malaysia (XMUM) AI Club**

**Submission Deadline: 25 October 2026**

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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**SDC HACKATHON 2026**

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## 1. Team & Project Overview

| | |
|---|---|
| **Team Name** | ONE Third Wheel |
| **Selected Domain** | Domain 2 |
| **Selected SDG Track (1–17)** | SDG 8: Decent Work and Economic Growth |
| **Project / Solution Name** | TeamSync |
| **Name (Leader)** | Hussein Al-Manifi |
| **Email & Phone (Leader)** | halmanifi@gmail.com +6011 4008 8061 |
| **University Affiliation(s)** | University of Malaya |
| **Number of Team Members** | 3 |

## 2. Executive Summary & Problem Statement

### 2.1 The Problem Context

*Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.*

***[Maximum 250 words]***

Across Malaysian SMEs, work is coordinated in chat. Tasks, deadlines and decisions are agreed in group chats, meeting notes and private messages, but nothing turns those conversations into a shared, reliable record of who is doing what.

This creates a hidden productivity gap. Commitments get buried in message history and forgotten. Employees repeat the same progress update to different people. Team members cannot see what colleagues are working on, so work overlaps or stalls. Managers must chase updates before they can spot blockers or redistribute work, and cross-team projects have no single, current view of responsibilities.

Dedicated project-management tools exist, but they demand manual entry, training and new habits, so small teams often struggle to sustain them and fall back on chat and memory.

This matters because SMEs are central to Malaysia's economy. In 2024, MSMEs contributed 39.5% of GDP and employed 48.7% of the workforce, about 8.1 million people, yet their labour productivity grew only 2.6%. Hours lost to status reporting and missed commitments, repeated across millions of workers, hold back that productivity, and with it wages, competitiveness and growth.

Those affected are employees carrying an invisible coordination burden, managers making decisions without visibility, and owners whose businesses struggle to grow beyond informal processes. This connects to SDG 8, specifically higher productivity through technology (Target 8.2) and the growth of SMEs (Target 8.3).

### 2.2 The Proposed AI Solution

*Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.*

***[Maximum 300 words]***

TeamSync is a chat application with a built-in AI assistant that turns workplace conversations into a shared, reliable view of who is doing what. Teams keep talking where they already work; TeamSync handles the tracking.

When a group admin enables TeamSync, the AI reads new messages in that group and identifies agreed actions, owners and deadlines. It distinguishes real commitments from questions, suggestions or hypotheticals, and never guesses a missing owner or deadline; it asks for clarification instead. Each suggestion is a proposal: a person confirms it before it becomes a task, so humans stay in control. In private chats, nothing is analysed automatically. Users highlight specific messages and select “Jot this down” to capture a commitment.

Every confirmed task is a single record that appears in three places: the group’s overview, the owner’s personal task list, and their team’s task list, where teammates and the manager can see progress, blockers and deadlines without asking for updates. Updating a task once updates it everywhere. Seeing a task never reveals the conversation it came from, so private discussions stay private.

We will apply skills from the SDC Phase 1 workshops across the stack:

- **GenAI APIs:** a large language model returns structured task proposals, validated by the backend before anyone sees them. The AI has no direct write access.
- **Database architecture:** a PostgreSQL schema with one canonical task table, role-based permissions enforced on every query, and an audit trail of every change.
- **Cloud hosting:** a React and TypeScript frontend with a FastAPI backend, deployed on Replit with real-time chat over WebSockets.

The result is structured, accountable task tracking that requires no extra data entry, designed for SMEs that run on chat.

## 3. Technical Architecture & Implementation Plan

### 3.1 Conceptual Tech Stack

*List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.*

*Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.*

The architecture has four layers, connected top to bottom.

1. **Users:** Employees and managers access TeamSync through a web browser.
2. **Cloud hosting (Replit):** Everything inside this layer runs on a single always-on Replit deployment.
    - The React and TypeScript frontend provides the chat interface, each group's TeamSync overview, and the My tasks and Team tasks tabs. It talks to the backend through a REST API for data and WebSockets for live chat and task updates.
    - The FastAPI backend is the only component that reads or writes data. It enforces all permissions, runs the approval workflow, and hands chat messages to the AI task extractor.
    - The AI task extractor runs as a background job inside FastAPI. It sends messages from TeamSync-enabled groups, or messages a user selected with "Jot this down", to our open-source language model on RunPod.
3. **Database architecture (PostgreSQL):** Hosted on Replit alongside the application, it stores users, teams, conversations, messages, tasks, AI proposals and an audit log. Each task is one record shown in every authorised view.
4. **AI hosting (RunPod):** An open-source large language model runs on a RunPod GPU and is exposed as an API, which the backend calls from Replit. It returns structured task proposals (title, owner, deadline, status). The backend validates each one and saves it as a proposal; it becomes a task only after a person confirms it. The AI never writes to the database directly.

Refer to Appendix 8.1 for the system block diagram

### 3.2 Design-Thinking & Deployment Strategy

*Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.*

***[Maximum 200 words]***

We applied the five design-thinking stages to keep TeamSync grounded in how SME teams actually work.

**Empathise:** All three of us saw this communication gap first-hand during our internships: tasks agreed in chats and meetings were easily lost, and progress had to be repeated to different people. We turned these experiences into personas such as Emma, a team member, and her manager, David.

**Define:** We framed the core problem as the repeated effort of turning scattered conversations into reliable work status.

**Ideate:** Instead of another project-management app, we chose to work inside chat, with human confirmation of every AI suggestion, opt-in analysis per group, and private-chat capture only for selected messages.

**Prototype:** We built and tested the core first: one task shown in group, personal and team views, with permission tests confirming correct access.

**Test:** We will measure time spent capturing tasks, correcting AI suggestions and reporting status against the current process.

**Phase 2:** The app will run on an always-on Replit deployment, with the open-source model on RunPod GPUs, backend validation, clear failure handling and an audit trail. We will extend the same detect-and-confirm approach to SME admin requests raised in chat: automated invoices, leave applications, expense claims and calendar invites.

## 4. Preliminary Environmental & SDG Impact Framework

*To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.*

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess* ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

| **Metric Category** | **Your Response** |
|---|---|
| **Target SDG Sub-Target(s)**<br>*Identify the exact UN SDG sub-target your project addresses*<br>*e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."* | Target 8.2: "Achieve higher levels of economic productivity through diversification, technological upgrading and innovation."<br><br>Supporting: Target 8.3: promote policies that support productive activities and the formalisation and growth of micro, small and medium-sized enterprises.<br><br>TeamSync applies AI to raise SME workforce productivity by automating the capture and tracking of work commitments made in chat. |
| **Current Problem Baseline**<br>*What is the starting condition of the problem, expressed as a number?*<br>*e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."* | Knowledge workers spend 60% of their time on "work about work", such as status updates and chasing information, instead of skilled work, and 26% of deadlines are missed each week (Asana Anatomy of Work Index 2021, 13,123 workers across 7 countries including Singapore). Workers also spend about 25% of their week searching for information (Atlassian State of Teams 2025).<br><br>This matters at scale in Malaysia: MSMEs employ 48.7% of the workforce, yet labour productivity grew only 2.6% in 2024 (DOSM, 2025). |
| **Core Measurable KPI**<br>*The specific unit you will track* | **Primary:** Minutes per task spent capturing a commitment from chat and reporting its status, comparing the manual process with TeamSync. |

| Metric Category | Your Response |
|---|---|
| *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.* | **Supporting:** % of agreed commitments in chat captured as tasks with the correct owner and deadline (capture accuracy).<br><br>Measured through timed trials with 5–10 participants using scripted group and private chats containing about 50 seeded commitments plus decoy messages. |
| **Phase 2 Impact Target**<br>*Your realistic, percentage-based target by the final prototype presentation*<br>*e.g., "20% improvement in computational resource use."* | **≥ 50% reduction** in time spent capturing and reporting task status per task, compared with the manual process.<br><br>Supported by **≥ 85% capture accuracy**, with no more than 10% of AI suggestions rejected as incorrect. |

## 5. Eligibility & Readiness Checklist

*Tick all boxes that apply before submitting.*

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

## 6. IP & Originality Declaration

### 6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

### 6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

| Name & Version | Purpose in Project | License |
|---|---|---|
| Python & 3.12 | Backend | PSF License (Python Software Foundation License) |
| React & 19.1 | Frontend | MIT |
| FastAPI & 0.115 | Backend (API) | MIT |
| TypeScript & 5.8.3 | Frontend | Apache-2.0 |
| Sqlalchemy & 2 | Backend (DB) | MIT |
| Psycopg & 3.2 | Backend (DB) | LGPL-3.0 |
| Alembic & 1.13 | Backend (DB) | MIT |
| Pillow & 13 | Backend | MIT-CMU (HPND-style) |
| Email-validator & 2.3 | Backend | Unlicense (public domain) |

| Name & Version | Purpose in Project | License |
|---|---|---|
| Python-multipart & 0.0.32 | Backend | Apache-2.0 |
| Pyyaml & 6.0.3 | Backend (YML) | MIT |
| Argon2-cffi & 25.1 | Backend (Password Hashing) | MIT |
| Uvicorn & 0.3 | Backend (ASGI) | BSD-3-Clause |
| Httpx & 0.27 | Backend (API) | BSD-3-Clause |
| Ruff & 0.6 | Linter | MIT |
| Openapi-spec-validator & 0.9 | Backend | Apache-2.0 |
| Datamodel-code-generator & 0.86 | Backend | MIT |
| Jsonschema & 4.26 | Backend | MIT |
| Openapi-typescript & 7.13 | Frontend (Converter) | MIT |
| Vite & 6.3.5 | Frontend | MIT |
| Claude | Code Generator, Project Helper | Proprietary. Governed by Anthropic's Consumer Terms or Commercial Terms. Claude Code is also proprietary. |
| Codex | Code Generator | Apache-2.0. The hosted Codex cloud agent and models are proprietary, under OpenAI's terms. |
| ChatGPT | Rephraser | Proprietary. Governed by OpenAI's Terms of Use and Services Agreement. |
| Replit | Code Generation / Design | Proprietary. Governed by Replit's Terms of Service. Some of its underlying tooling is open source, but the platform and its AI agent are not. |
| Qwen3-8B | Task Suggestor | Apache-2.0 |

### 6.3 Employment-Related IP Conflicts

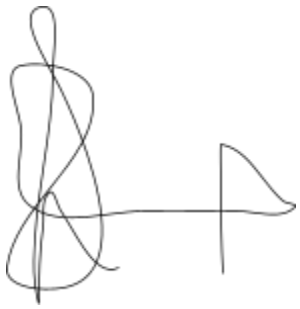
Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No
✓

If yes, please describe the nature of the agreement and which member it concerns:

## 7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



24 Oct 2026

# 8. Appendix

## 8.1 TeamSync System Architecture

